

PRATIK PATTANAYAK

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SUMMARY

Results-driven Backend Engineer and AWS-certified Machine Learning professional with expertise in architecting scalable backend systems, developing advanced analytical APIs, and seamlessly integrating Generative AI (LLMs) into production environments. Proven ability to engineer automated workflows, optimize application metrics, and deploy reliable AI/ML software solutions utilizing Python, JavaScript, and modern frameworks

PROFESSIONAL EXPERIENCE

Backend Engineer | Dirtcube Interactive (June 2026 – Present)

- **Backend Analytics & APIs:** Built an extensive suite of analytical UI routes, backend API endpoints, and data visualization tools for a comprehensive digital workspace dashboard.
- **System Architecture & Notifications:** Engineered and managed the implementation of an Outreach Engine, developing complex database schema migrations and a segment resolver for a new push notification system.
- **Performance Optimization & Integration:** Optimized SQL queries to improve application metrics, configured frontend funnel components, and contributed to a platform featuring integrated AI agents managing marketing campaigns.

Self-Employed / Freelance | Python & AI Developer (2024 – Present)

- **End-to-End AI Engineering:** Architected, developed, and deployed custom AI-powered applications, intelligent desktop assistants, and network automation tools.
- **Backend & LLM Architecture:** Integrated Generative AI APIs (Gemini, LLaMA) into production workflows, applying advanced Prompt Engineering and Context Building techniques to optimize LLM responses and reasoning capabilities.

TECHNICAL SKILLS

- **Coding Languages:** Python (Advanced), Java, JavaScript, TypeScript, Flutter, Dart, C#, C++
- **AI & Generative AI:** Google Gemini 1.5 API, Groq API (LLaMA), Prompt Engineering, AI Agent/Workflow Design, NLP, ChromaDB, RAG, Vector Search.
- **ML Concepts:** Predictive Modeling, Data Preprocessing, Scikit-learn, Pandas, Model Evaluation, Fine Tuning, TensorFlow.
- **Backend & Automation:** Flask, RESTful APIs, Socket Programming, WhatsApp Automation, Gmail (IMAP/SMTP).
- **Databases, Frontend & Frameworks:** MySQL, PostgreSQL, SQLite, React, PyQt5/Tkinter.
- **Tools & Deployment:** Git, GitHub, VS Code, Postman, AWS (Cloud Fundamentals).

TECHNICAL PROJECTS

Cypher | AI-Powered Workflow & Desktop Automation Agent (Personal Project)

- Engineered a fast, multi-model AI assistant converting natural language into system-level automation.
- **Generative AI & Semantic Understanding:** Integrated Gemini 1.5 Flash as the primary reasoning engine, leveraging context management and semantic parsing to accurately convert natural language into reliable system-level automation.
- **Automated Workflows:** Built agentic workflows utilizing Python's smtplib/imaplib for autonomous Gmail reading/sending, and pywhatkit for targeted WhatsApp message dispatch.

PyShare | High-Speed Local Data Transfer Platform (Personal Project)

- Built a custom backend networking tool to handle seamless data processing and file transfers.
- **Backend Logic:** Utilized pure Python Socket Programming and HTTP protocols to bypass external servers, enabling instant, high-speed offline PC-to-Mobile data synchronization.

EDUCATION & CERTIFICATIONS

- **B.Sc. Information Technology – Graduated 2026 (Overall CGPI: 8.65 | Semester 6 SGPA: 9.10)**
- **AWS Certified:** Fundamentals of Machine Learning and Artificial Intelligence
- **Certifications:** Python for Data Science (SkillEcted) | MS-CIT Certification (MKCL, 2024)
- **Academics:** Higher Secondary (WBCHSE) – 65.2% (2023) | Secondary (WBBSE) – 83.85% (2021)